ME 351 Feedback Control Systems

Root Locus Method

Dirk M. Luchtenburg | Fall 2026

Idea

The *root locus* (Evans) shows how the **closed-loop poles** move as a real gain K varies from

$$\boxed{0 \leq K < \infty}.$$

For a standard negative-feedback system, write the closed-loop characteristic equation in *Evans form*

$$\boxed{1 + KL(s) = 0, \qquad L(s) = \frac{n(s)}{d(s)}.} \tag{1}$$

Equivalently,

$$d(s) + Kn(s) = 0.$$

The root locus is simply the set of all s satisfying this equation as K varies.

Magnitude and Phase Conditions

From

$$1 + KL(s) = 0$$

we obtain

$$KL(s) = -1.$$

Since s is generally complex, both *magnitude* and *phase* must agree.

For $K > 0$,

$$\boxed{K|L(s)| = 1}$$

and

$$\boxed{\angle L(s) = (2\ell + 1)\pi, \qquad \ell \in \mathbb{Z}.}$$

Thus, a point s lies on the root locus if it satisfies the *angle condition*. Once such a point is found, the corresponding gain follows from the *magnitude condition*:

$$\boxed{K = \frac{1}{|L(s)|}.} \tag{2}$$

If

$$L(s) = \frac{\prod_{j=1}^{m}(s - z_j)}{\prod_{i=1}^{n}(s - p_i)},$$

then the angle condition can be written

$$\boxed{\sum_{j=1}^{m} \angle(s - z_j) - \sum_{i=1}^{n} \angle(s - p_i) = (2\ell + 1)\pi.} \tag{3}$$

This equation is the basis of the root-locus construction rules.

Minimal Root-Locus Rules

Assume $L(s)$ has n open-loop poles and m finite open-loop zeros, with

$$n \geq m.$$

There are n root-locus branches.

0. Symmetry

For systems with real coefficients, complex poles and zeros occur in conjugate pairs. Therefore,

$$\boxed{\text{The root locus is symmetric about the real axis.}} \tag{4}$$

It is therefore sufficient to determine one half of the locus.

1. Start and End Points

The characteristic equation is

$$d(s) + Kn(s) = 0.$$

At $K = 0$,

$$d(s) = 0.$$

Therefore, every root-locus branch *starts* at an open-loop pole:

$$\boxed{K = 0 : \qquad s = p_i.} \tag{5}$$

As $K \to \infty$, divide the characteristic equation by K:

$$\frac{1}{K} d(s) + n(s) = 0.$$

For finite s,

$$n(s) = 0.$$

Thus, branches *end* at open-loop zeros:

$$\boxed{K \to \infty : \qquad s \to z_j.} \tag{6}$$

If there are fewer zeros than poles, $m < n$, only m branches can end at finite zeros. The remaining

$$\boxed{r = n - m}$$

branches go to infinity.

The integer r is the *relative degree* of $L(s)$.

Hence,

branches start at open-loop poles and end at open-loop zeros or at infinity.

2. Real-Axis Segments

Consider a candidate point s on the real axis. The root-locus phase condition requires

$$\angle L(s) = \sum_j \angle(s - z_j) - \sum_i \angle(s - p_i) = (2\ell + 1)\pi.$$

For a real pole or zero, the angle seen from s is particularly simple:

$$\begin{aligned} \text{pole or zero to the left of } s &\implies \angle(s - p_i),\ \angle(s - z_j) = 0, \\ \text{pole or zero to the right of } s &\implies \angle(s - p_i),\ \angle(s - z_j) = \pi \pmod{2\pi}. \end{aligned}$$

Thus, only the real poles and zeros *to the right* of the candidate point contribute to the phase condition. Each contributes a factor of π in phase (the sign is irrelevant modulo 2π).

If there are N_R real poles and zeros to the right of s, then

$$\angle L(s) = N_R \pi \pmod{2\pi}.$$

The Evans phase condition requires an odd multiple of π, so

$$N_R = 1, 3, 5, \ldots$$

Therefore,

A point on the real axis lies on the root locus if the number of real open-loop poles and zeros to its right is odd.

(7)

This rule is therefore not an additional construction rule: it is simply the Evans phase condition specialized to the real axis.

3. Asymptotes: What Happens at Infinity?

Suppose

$$L(s) = \frac{n(s)}{d(s)} = \frac{\prod_{j=1}^{m}(s - z_j)}{\prod_{i=1}^{n}(s - p_i)}, \qquad n > m,$$

and define the *relative degree*

$$\boxed{r = n - m.}$$

There are n root-locus branches. Since only m branches can terminate at finite zeros, the remaining $r = n - m$ branches must go to infinity.

To determine their behavior, consider $|s|$ much larger than the magnitude of every finite pole and zero. Then

$$s - p_i \approx s, \qquad s - z_j \approx s,$$

and therefore

$$L(s) \approx \frac{s^m}{s^n} = \boxed{\frac{1}{s^r}}.$$

More generally, if the leading coefficients of $n(s)$ and $d(s)$ are b_m and a_n,

$$L(s) \sim \frac{b_m}{a_n}\frac{1}{s^r}.$$

Thus, far from all finite poles and zeros, the detailed pole–zero locations become unimportant: the behavior is determined by the *excess number of poles over zeros.*

At large $|s|$, relative degree determines the root-locus geometry.

Asymptote angles

Write

$$s = \rho e^{j\theta}.$$

For large ρ,

$$L(s) \sim \frac{1}{s^r} = \frac{1}{\rho^r}e^{-jr\theta},$$

so

$$\angle L(s) \approx -r\theta.$$

The Evans phase condition is

$$\angle L(s) = (2\ell + 1)\pi.$$

Therefore,

$$-r\theta = (2\ell + 1)\pi \qquad (\text{mod } 2\pi).$$

The r distinct directions are

$$\boxed{\theta_q = \frac{(2q+1)\pi}{r}, \qquad q = 0, 1, \ldots, r-1.} \tag{8}$$

Thus, the familiar asymptote-angle rule is simply the *root-locus phase condition applied as* $|s| \to \infty$.

For example, if $r = 3$,

$$L(s) \sim \frac{1}{s^3},$$

and the three asymptotic directions are

$$\boxed{\theta = 60^\circ, \quad 180^\circ, \quad 300^\circ.}$$

Where do the asymptotes intersect?

The approximation $L(s) \sim 1/s^r$ gives the asymptote directions. To determine their intersection on the real axis, retain one more term in the large-s behavior.

Write

$$n(s) = s^m - \left(\sum_j z_j \right) s^{m-1} + \cdots ,$$

and

$$d(s) = s^n - \left(\sum_i p_i \right) s^{n-1} + \cdots .$$

For large s,

$$L(s) = \frac{n(s)}{d(s)} \sim \frac{1}{s^r} \left[1 + \frac{\sum_i p_i - \sum_j z_j}{s} + \cdots \right] .$$

Now compare this with

$$\frac{1}{(s-\alpha)^r}.$$

For large s,

$$\frac{1}{(s-\alpha)^r} = \frac{1}{s^r} \left(1 - \frac{\alpha}{s} \right)^{-r} \sim \frac{1}{s^r} \left(1 + \frac{r\alpha}{s} + \cdots \right) .$$

Matching the $1/s$ terms gives

$$r\alpha = \sum_i p_i - \sum_j z_j .$$

Hence,

$$\boxed{\alpha = \frac{\sum_i p_i - \sum_j z_j}{r} .} \tag{9}$$

The r branches going to infinity therefore approach straight lines passing through the centroid α at the angles θ_q.

r	asymptote angles
1	180°
2	90°, 270°
3	60°, 180°, 300°
4	45°, 135°, 225°, 315°

Sign Matters

The preceding rules assume the Evans standard form

$$\boxed{1 + KL(s) = 0, \qquad K \geq 0.}$$

The phase condition then follows from

$$KL(s) = -1,$$

so that

$$\angle L(s) = (2\ell + 1)\pi.$$

A minus sign changes the phase by π. Therefore, signs must not be discarded when forming $L(s)$.

For example, if the characteristic equation is instead written

$$1 - KL(s) = 0, \qquad K \geq 0,$$

then

$$KL(s) = 1,$$

and the phase condition becomes

$$\boxed{\angle L(s) = 2\ell\pi.}$$

Consequently, the usual odd-number real-axis rule and asymptote angles also change.

Best practice: first put the characteristic equation explicitly into the chosen Evans form and then apply the corresponding phase condition.

Root Locus as a Design Tool

The root locus connects the adjustable gain directly to closed-loop pole locations:

$$\boxed{K \quad \longrightarrow \quad \text{closed-loop poles} \quad \longrightarrow \quad \text{transient response.}}$$

For a dominant complex pole pair,

$$s = -\zeta\omega_n \pm j\omega_d,$$

pole location determines damping, overshoot, settling time, and oscillation frequency.

The root locus therefore answers the design question:

$$\boxed{\text{As I change } K, \text{ where can the closed-loop poles go?}}$$

Once a desired point $s^\star$ on the locus has been selected from performance requirements, the required gain follows immediately from

$$\boxed{K = \frac{1}{|L(s^\star)|}.}$$

Summary

For

$$1 + KL(s) = 0, \qquad L(s) = \frac{n(s)}{d(s)}, \qquad K \geq 0,$$

the essential facts are

Phase:	$\angle L(s) = (2\ell + 1)\pi$,
Magnitude:	$K = \dfrac{1}{\lvert L(s) \rvert}$,
Start:	open-loop poles,
End:	open-loop zeros or infinity,
Real axis:	odd number of real poles/zeros to the right,
Relative degree:	$r = n - m$,
Centroid:	$\alpha = \dfrac{\sum p_i - \sum z_j}{r}$,
Asymptote angles:	$\theta_q = \dfrac{(2q + 1)\pi}{r}$.

These rules give the essential geometry. More specialized rules—such as breakaway points and angles of departure—can be derived from the same magnitude and phase conditions when needed.